David Hristov

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Personal Website
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Summary

Computer Science and Engineering graduate specializing in machine learning and intelligent systems, with hands-on experience across traditional ML, deep learning, natural language processing, and agentic AI systems. Interested in designing, training, and deploying reliable AI solutions as part of an engineering team.

Education

MSc in Data Science in Computer Science and Engineering *November 2026 – Present*
Ss. Cyril and Methodius University in Skopje, Faculty of Computer Science and Engineering (FCSE)

BSc in Computer Science and Engineering *1 October 2022 – 6 October 2026*
Ss. Cyril and Methodius University in Skopje, Faculty of Computer Science and Engineering (FCSE)
Thesis: Comparative Analysis over Knowledge Graphs Based on Multi-Hop Reasoning through Path Discovery and Evaluation
Relevant coursework: Machine Learning, Natural Language Processing, Intro to Data Science, Mining Massive Datasets, Web-Based Systems, Service-Oriented Architectures, DevOps (CI/CD)

Experience

Machine Learning Intern
Loka *(May 2026 – September 2026)*
- Completed a comprehensive 4-month program building end-to-end ML systems and cutting-edge LLM applications, including transformer implementation, supervised fine-tuning, RL and multi-agent RAG systems.
- Contributed to a client production project using multimodal LLMs on AWS Bedrock, benchmarked models on accuracy, reliability and cost.
- Gained hands-on MLOps experience with AWS services (Lambda, S3, EC2, Bedrock, IAM) and Infrastructure as Code (AWS CDK) under mentorship of seasoned DevOps engineers

Software Engineering Intern
Inteligenta *(April 2025 – August 2025)*
- Contributed to enterprise-scale system development within an agile environment participating in sprint planning, daily stand-ups, and retrospectives.
- Developed full-stack features and integration layers, improving service communication and data workflows through clean design practices and SOLID principles.
- Refactored legacy components to enhance reliability, maintainability, and system performance.

Publications

- **D. Hristov**, S. Tagarski, D. Sasanski, R. Stojanov. **"VisualKGQA: Multi-Hop Reasoning Path Visualization."** *International Conference on Informatics and Information Technologies (CIIT 2026).* Selected for IEEE Xplore publication (approx. 30% of accepted papers) to appear.
- S. Tagarski, **D. Hristov**, M. Poposki, S. Andonov, G. Madjarov. **"Real-Time Cascading Failure Detection in Smart Grid Networks Using Adaptive Stream Processing."** Under review, (Telfor 2026, Belgrade).

Notable Projects

More of my projects and additional information available on my website.

Multi-tier Router - A Compliance-Aware Cost Optimizer for LLMs
- Built a multi-tier LLM query router on AWS that sends each query to a deterministic tier, fast tier, a strong tier or an isolated private tier, using Titan embeddings with S3 Vectors for route matching.
- Deployed the private tier for sensitive queries on ECS Fargate in an isolated subnet with no internet route, running local embeddings, FAISS retrieval, and a local Gemma model, reachable only through VPC endpoints.
- Set up shared RAG for the cloud tiers with a Bedrock Knowledge Base backed by S3 Vectors.
- Built an evaluation pipeline with Step Functions (Distributed Map) and Lambda that runs benchmark queries across tiers and uses an LLM judge to find the best tier per query, used to assess routing decisions.

Cross-Check – Multi-Agent Research Verifier

- Designed a multi-agent system that verifies technical claims by cross-checking the paper knowledge base against live web search, with a lead agent drafting an evidence-backed verdict.
- Built a RAG pipeline over a corpus of research papers, evaluating retrieval recall and showing that grounding reduces hallucinations, making the model admit missing information instead of guessing.
- Implemented a critic agent that rejects unsupported claims and sends drafts back for revision, plus a debate round between agents whenever the sources disagree.
- Added persistent memory of past verdicts, full observability with Langfuse tracing, and a live UI that streams each agent's progress.

Multi-Topic LLM Fine-Tuning with LoRA

- Fine-tuned Qwen2.5-1.5B on GSM8K math reasoning, running rank ablations, baseline comparisons, and catastrophic-forgetting checks.
- Built an LLM-as-judge evaluation pipeline with bias detection to score fine-tuned models beyond exact-match accuracy.
- Trained a single LoRA adapter jointly across four domains (math, code, logic, biology) to test whether one adapter can replace separate per-domain specialists, raising code accuracy from 0% to 52% and improving logic and biology by 6–8 points.
- Identified negative transfer on math reasoning (about a 10-point drop) and cut training time from 4 hours to 1 hour per run by restructuring gradient accumulation.

Technical Skills

- **ML & Deep Learning:** PyTorch, TensorFlow/Keras, scikit-learn, NumPy, Pandas, XGBoost, Hugging Face Transformers (BERT), CNNs, transfer learning, LoRA fine-tuning on open-weight LLMs (PEFT, Unsloth), hyperparameter tuning, feature engineering, model explainability (Grad-CAM)
- **LLMs & NLP:** RAG (LangChain), single and multi-agent architectures (Strands Agents, LangGraph), MCP (FastMCP), tool calling, structured outputs (Pydantic), LLM-as-judge evaluation, embeddings, Sentence Transformers, spaCy
- **Databases & Vector Stores:** PostgreSQL, MySQL, MongoDB, FAISS, pgvector, ChromaDB, S3 Vectors
- **Cloud & Infrastructure:** AWS (S3, Lambda, EC2, ECS/Fargate, ECR, IAM, SageMaker, Bedrock, Bedrock AgentCore), AWS CDK, Azure (App Service, AI Foundry, Azure OpenAI, AI Search, Cosmos DB)
- **DevOps & MLOps:** Docker, Docker Compose, Kubernetes (k3d), CI/CD (GitHub Actions, Jenkins), MLflow, Langfuse
- **Data Engineering:** Apache Spark/PySpark (batch processing), Apache Kafka (event streaming), Apache Flink (stream processing), event-driven pipelines
- **Backend Development:** FastAPI, Spring Boot, ASP.NET Core, RESTful APIs, microservices, CQRS, event-driven architecture
- **Tools:** Git, Unix/Linux, Streamlit, React, Postman, CUDA

Certifications

- Deep Learning with TensorFlow & PyTorch
- Latency-limited processing of streaming data with Apache Flink
- Natural Language Processing
- Day After Production - Cybersecurity and Exploiting

Languages

- Macedonian: Native
- English: Fluent